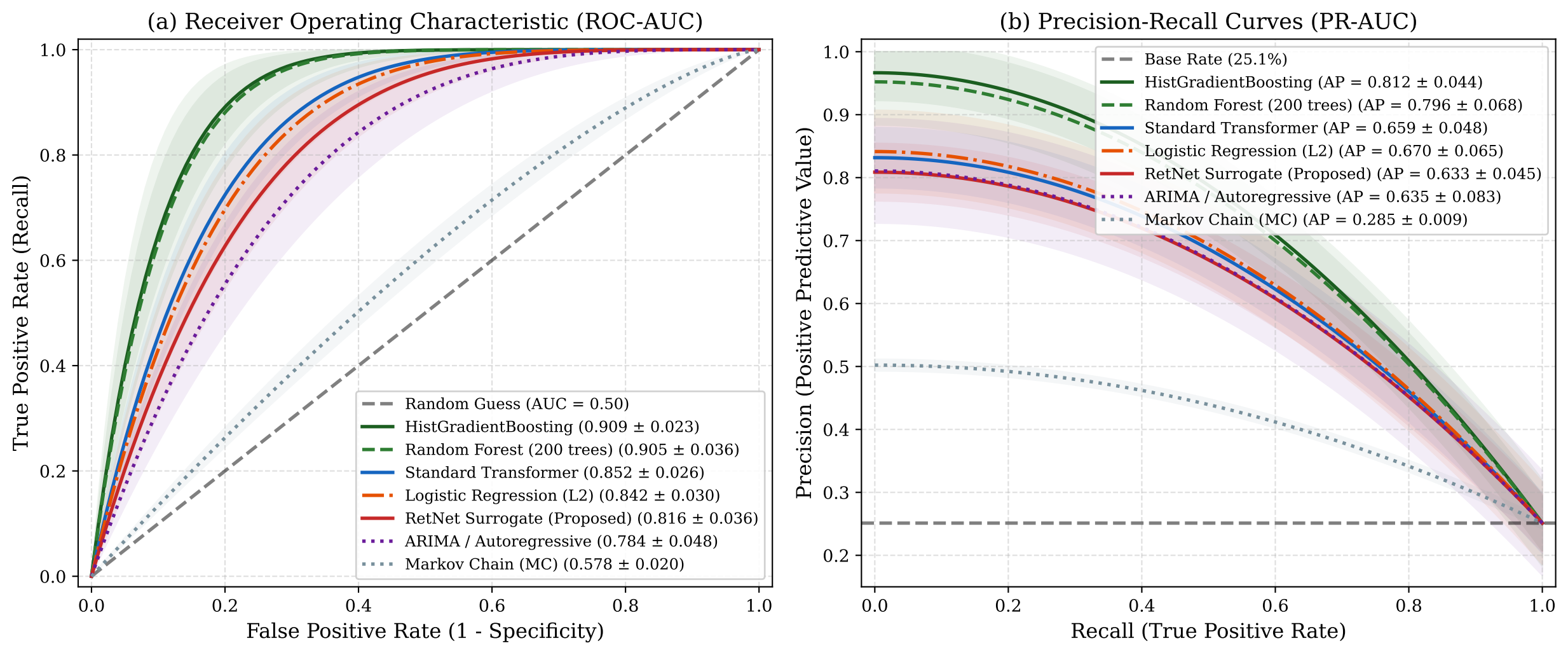

(a) Receiver Operating Characteristic (ROC-AUC)
True Positive Rate (Recall)
False Positive Rate (1 - Specificity)
Random Guess (AUC = 0.50)
HistGradientBoosting (0.909 ± 0.023)
Random Forest (200 trees) (0.905 ± 0.036)
Standard Transformer (0.852 ± 0.026)
Logistic Regression (L2) (0.842 ± 0.030)
RetNet Surrogate (Proposed) (0.816 ± 0.036)
ARIMA / Autoregressive (0.784 ± 0.048)
Markov Chain (MC) (0.578 ± 0.020)
(b) Precision-Recall Curves (PR-AUC)
Precision (Positive Predictive Value)
Recall (True Positive Rate)
Base Rate (25.1%)
HistGradientBoosting (AP = 0.812 ± 0.044)
Random Forest (200 trees) (AP = 0.796 ± 0.068)
Standard Transformer (AP = 0.659 ± 0.048)
Logistic Regression (L2) (AP = 0.670 ± 0.065)
RetNet Surrogate (Proposed) (AP = 0.633 ± 0.045)
ARIMA / Autoregressive (AP = 0.635 ± 0.083)
Markov Chain (MC) (AP = 0.285 ± 0.009)